AP PRECALCULUS

Unit 1 • Worksheet 1.7A • Answer key

Worksheet 1.7A • Rational Functions and End Behavior

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the horizontal asymptote of $r(x) = \dfrac{5x - 1}{x^2 + 4}$?

(A) $y = 5$
(B) $y = 0$
(C) $y = -\frac{1}{4}$
(D) There is none.

Solution

Answer: (B). Degree of numerator $<$ degree of denominator.

Question 2

NO CALCULATOR

What is the horizontal asymptote of $r(x) = \dfrac{4x^3 - x}{1 - 2x^3}$?

(A) $y = 2$
(B) $y = 4$
(C) $y = -2$
(D) $y = 0$

Solution

Answer: (C). $\frac{4}{-2} = -2$.

Question 3

NO CALCULATOR

What is the slant asymptote of $r(x) = \dfrac{x^2 + 3x}{x - 1}$?

(A) $y = x + 3$
(B) $y = x + 4$
(C) $y = x$
(D) $y = x - 4$

Solution

Answer: (B). $x^2 + 3x = (x - 1)(x + 4) + 4$.

Question 4

NO CALCULATOR

$\lim_{x\to-\infty} \dfrac{x^3}{x^2+1}$ is

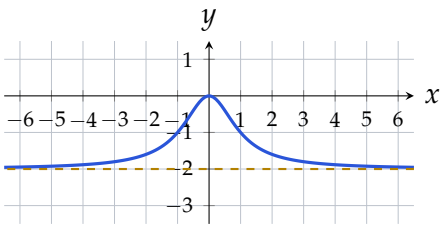
(A) 1
(B) 0
(C) ∞
(D) $-\infty$

Solution
Answer: (D). Behaves like x.

Question 5

NO CALCULATOR

The graph of $r(x) = \dfrac{-2x^2}{x^2+1}$ is shown. What is $\lim_{x\to\infty} r(x)$?



(A) 0
(B) -2
(C) $-\infty$
(D) 2

Solution
Answer: (B). The graph levels off at $y = -2$.

Question 6 (Free response)

NO CALCULATOR

Let $r(x) = \dfrac{6x^2+x}{3x^2-12}$ and $s(x) = \dfrac{2x^2-x+3}{x+1}$.

(a) Write limit statements for the end behavior of r.
(b) Find the slant asymptote of s.
(c) For large positive x, is the graph of s above or below its slant asymptote? Explain.

Solution
(a) $\lim_{x\to\pm\infty} r(x) = 2$.
(b) $s(x) = 2x - 3 + \frac{6}{x+1}$: $y = 2x - 3$.
(c) $\frac{6}{x+1} > 0$ for $x > -1$: above.